

Gradient Descent

$$\rightarrow Z = Wx$$

$$J(W) = \frac{1}{2m} \sum_{i=1}^m (z_i - w x_i)^2$$

↙ (Cost function)

Best value of w ?

↓
Gradient Descent

$$\frac{\partial J}{\partial w}$$

But how to progress

$$w_0 \rightarrow w_1$$

25 ↗

$$w_0 \rightarrow w_1$$

$$\left[w_1 = w_0 - \eta \frac{\partial J}{\partial w} \right]$$

follow the repeat

~~w_0~~
 ~~w_1~~

Back propagation

Calculate E
 Travel to the initial
 network
 (update w)

$$\rightarrow w_2 \rightarrow w_2 - \eta \frac{\partial E}{\partial w_2}$$

$$E = \frac{1}{2} (T - a_2)^2 \rightarrow \frac{\partial E}{\partial a_2}$$

$$a_2 = f(z_2) = \frac{1}{1 + e^{-z_2}} \rightarrow \frac{\partial a_2}{\partial z_2}$$

$$z_2 = T - a_1 \quad 1 - e^{-z_2} \quad \frac{\partial z_2}{\partial w_2}$$

$$z_2 = a_1 \cdot w_2 + b_2 \rightarrow \frac{\partial z_2}{\partial w_2}$$



So, Backpropagation,

$$w_2 \rightarrow w_2 - \eta \left(-(T - a_2) \cdot (a_2(1 - a_2))(a_1) \right)$$

$$w_1 \rightarrow w_1 - \eta \frac{\partial E}{\partial w_1}$$



Similarly Back-prop.

also, update Biases b_2 and b_1

Vanishing Gradient

Vanishing Gradient

↳ The problem with activation fn when it prevent Neural Networks from Booming sooner is known as the vanishing gradient problem.

Activation Functions :

Types of Activation Function



- | | |
|----------------|------------|
| i) Binary step | ii) Linear |
| iii) Sigmoid | iv) Tanh |

- iii) Sigmoid iv) Tanh
- v) ReLU vi) Leaky ReLU
- vii) Softmax

• Sigmoid

$$a = \frac{1}{1 + e^{-z}}$$

• Tanh :

$$a = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

• ReLU :
(Non
linear)

$$a = \max(0, z)$$

↓
Sparse
Network
↓
Fixed

Vanishing
Gradient
Problem

• Softmax :

$$a_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

- Softmax :
$$a_i = \frac{e^{z_i}}{\sum_{k=1}^m e^{z_k}}$$